

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Dashboard Development Task

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO5: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication.

Assessment Tool Weightage: 40%

Total Marks: 20

Code of Conduct

I, _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Dashboard Development Task Rubric

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Data Importing, Exploration & Preparation	20	Thoroughly imports, explores, and prepares the dataset; correctly identifies data types, missing values, and relevant variables	Imports and explores data correctly with minor gaps in preparation or identification of issues	Performs basic data exploration and preparation but misses some important aspects	Unable to import or explore the dataset correctly; major preparation issues remain
Python Coding & Pandas/Matplotlib Implementation	25	Effectively applies Pandas and Matplotlib concepts; code is accurate, well-structured, and implements all required analyses	Applies appropriate Pandas and Matplotlib concepts with minor coding or implementation gaps	Limited application of relevant concepts; some required analyses are incomplete	Fails to apply relevant Python concepts; significant coding errors or incomplete implementation
Visualization & Plot Components	25	Selects highly appropriate charts; effectively uses titles, labels, legends, grids, formatting, subplots, and other plot options	Uses appropriate charts and plot components with minor formatting or presentation issues	Uses basic charts but some visualizations or plot components are inappropriate or incomplete	Uses inappropriate/missing visualizations; poor or unclear plot components
Dashboard Design & Analytical Interpretation	20	Develops a well-organized, coherent dashboard; clearly identifies meaningful	Dashboard is logically organized and identifies relevant	Dashboard has basic organization; identifies some patterns but	Dashboard is poorly organized or incomplete; patterns are incorrectly

ion		trends, relationships, distributions, and patterns with strong justification	patterns with adequate interpretation	interpretation is limited	identified or not interpreted
Output Presentation, Documentation & File Publishing	10	Clearly presents results; code is well documented, dashboard is professionally formatted, and output is correctly saved/published	Presents results clearly with minor documentation or output-formatting issues	Basic presentation and documentation; some outputs are unclear or incomplete	Poor presentation; inadequate documentation or failure to produce/save the required dashboard

Total Marks Awarded:

Signature of the Course Faculty

NAME:- S. THARUN KMALESH

REG NO:- 192473004

1. Student Performance Pattern Analysis

A college wants to analyze student academic performance using Python. You are provided with a dataset containing Student_ID, Department, Semester, Attendance, Assignment_Marks, Internal_Marks and Final_Marks. Develop a Python-based academic performance dashboard using Pandas and Matplotlib that imports and explores the dataset, displays average assessment marks, compares departments, shows average marks across semesters, examines attendance versus final marks, displays the final-mark distribution and performance categories, arranges the visualizations using subplots, and provides a short interpretation of the major patterns.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Student_ID":["S01","S02","S03","S04","S05","S06","S07","S08","S09","S10","S11","S12"],
    "Department":["CSE","CSE","ECE","ECE","IT","IT","CSE","ECE","IT","CSE","ECE","IT"],
    "Semester":[1,2,3,4,1,2,3,4,5,5,6,6],
    "Attendance":[72,80,85,90,76,88,84,92,81,95,87,91],
    "Assignment_Marks":[60,68,72,80,65,75,70,84,73,88,79,85],
    "Internal_Marks":[55,65,72,82,60,75,70,85,74,90,80,87],
    "Final_Marks":[58,68,76,88,63,78,73,91,77,94,84,92]})

print(df.head())
print(df.describe())

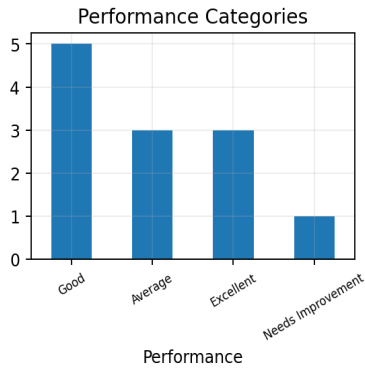
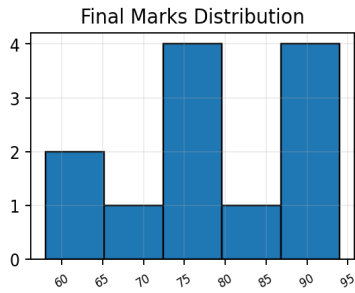
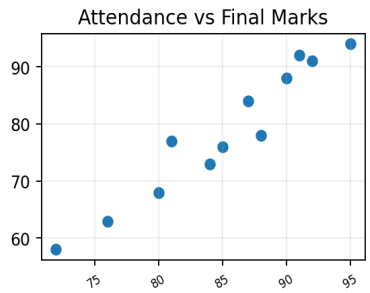
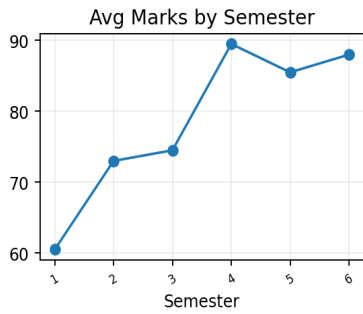
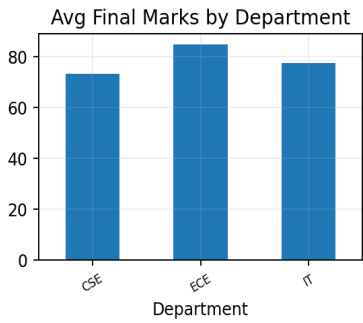
print("\nAverage assessment marks:")
print(df[["Assignment_Marks","Internal_Marks","Final_Marks"]].mean())

df["Performance"] = pd.cut(df["Final_Marks"],
    bins=[0,59,74,89,100],
    labels=["Needs Improvement", "Average", "Good", "Excellent"])

fig,ax=plt.subplots(2,3,figsize=(14,8))
df.groupby("Department")["Final_Marks"].mean().plot.bar(ax=ax[0,0],title="Average Final Marks by Department")
df.groupby("Semester")["Final_Marks"].mean().plot(ax=ax[0,1],marker="o",title="Average Marks by Semester")
ax[0,2].scatter(df["Attendance"],df["Final_Marks"])
ax[0,2].set(title="Attendance vs Final Marks",xlabel="Attendance",ylabel="Final Marks")
ax[1,0].hist(df["Final_Marks"],bins=5,edgecolor="black")
ax[1,0].set(title="Final Marks Distribution",xlabel="Final Marks",ylabel="Students")
df["Performance"].value_counts().plot.bar(ax=ax[1,1],title="Performance Categories")
ax[1,2].axis("off")
for a in ax.flat: a.grid(alpha=.25)
plt.tight_layout()
plt.savefig("student_dashboard.png",dpi=160)
plt.show()
```

Output

Google Colab Output — Student Academic Dashboard



Department comparison, semester trend, attendance relationship, final-mark distribution and performance categories are displayed. Pattern: higher attendance is associated with higher final marks. Pattern: CSE/ECE/IT show different average academic performance.

2. Sales and Customer Behavior Pattern Analysis

An e-commerce company wants to understand its sales performance using Date, Product_Category, Region, Quantity, Sales and Profit. Develop an analytical dashboard using Pandas and Matplotlib with sales over time, sales by category, profit by region, sales versus profit, sales distribution, order frequency by category, at least four subplots, suitable labels/annotations/gridlines, and at least three important sales patterns.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Date":pd.date_range("2026-01-01",periods=12,freq="MS"),
    "Product_Category":["Electronics","Clothing","Furniture","Electronics","Clothing","Furniture","Electronics","Clothing","Furniture","Electronics","Clothing","Furniture"],
    "Region":["North","South","East","West"]*3,
    "Quantity":[12,25,8,15,28,10,18,30,12,20,32,14],
    "Sales":[24000,15000,18000,30000,17500,22000,36000,19000,25000,41000,21000,28000],
    "Profit":[4800,3300,3600,6500,3850,4400,7200,4200,5000,8200,4600,5900]})

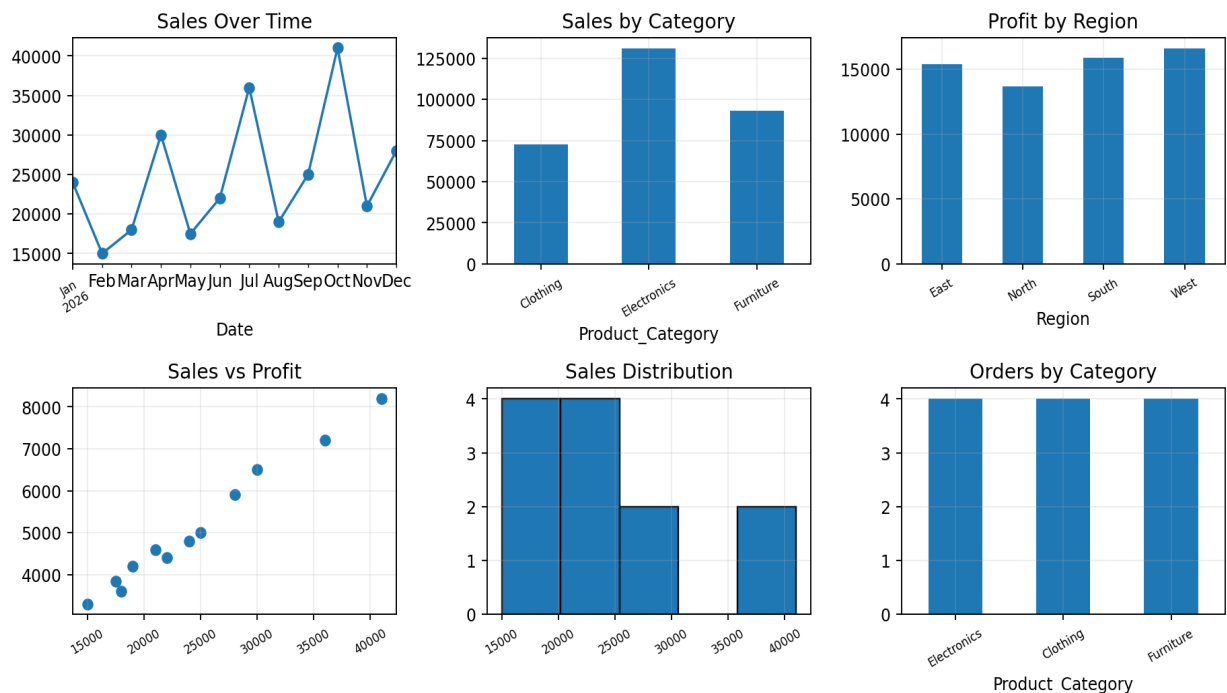
print(df.head())
print(df.describe())

fig,ax=plt.subplots(2,3,figsize=(14,8))
ax[0,0].plot(df["Date"],df["Sales"],marker="o",label="Sales")
ax[0,0].set(title="Sales Over Time",ylabel="Sales")
df.groupby("Product_Category")["Sales"].sum().plot.bar(ax=ax[0,1],title="Sales by Product Category")
df.groupby("Region")["Profit"].sum().plot.bar(ax=ax[0,2],title="Profit by Region")
ax[1,0].scatter(df["Sales"],df["Profit"])
ax[1,0].set(title="Sales vs Profit",xlabel="Sales",ylabel="Profit")
ax[1,1].hist(df["Sales"],bins=5,edgecolor="black")
ax[1,1].set(title="Sales Distribution",xlabel="Sales",ylabel="Orders")
df["Product_Category"].value_counts().plot.bar(ax=ax[1,2],title="Orders by Category")
for a in ax.flat:
    a.grid(alpha=.25)
    a.tick_params(axis="x",rotation=30)
plt.tight_layout()
plt.savefig("ecommerce_dashboard.png",dpi=160)
plt.show()

print("\nKey patterns:")
print("1. Electronics generates the highest sales in this sample.")
print("2. West region has the highest total profit.")
print("3. Sales and profit show a positive relationship.")
```

Output

Google Colab Output — E-Commerce Sales Dashboard



Electronics leads total sales in this sample.
West has the highest regional profit.
Sales and profit move positively.
Category order frequency is highest for Clothing.

3. Weather Data: Identifying Correlation and Temperature Patterns

An organization wants to analyze employee productivity using Employee_ID, Department, Experience, Working_Hours, Training_Hours, Projects_Completed and Productivity_Score. Develop a Productivity Analysis Dashboard that imports and explores the data, shows department counts, compares average productivity by department, plots working hours versus productivity, training hours versus productivity, productivity distribution, experience-group productivity, uses a dashboard layout, saves the dashboard with plt.savefig(), and summarizes the key patterns.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Employee_ID":["E01","E02","E03","E04","E05","E06","E07","E08","E09","E10","E11","E12"],
    "Department":["HR","IT","Sales","IT","HR","Sales","IT","Sales","HR","IT","Sales","HR"],
    "Experience":[1,3,5,8,2,10,6,4,12,7,9,11],
    "Working_Hours":[35,40,42,45,38,44,43,41,46,44,45,43],
    "Training_Hours":[10,15,20,25,12,30,22,18,35,24,28,32],
    "Projects_Completed":[2,4,6,8,3,10,7,5,11,8,9,10],
    "Productivity_Score":[55,68,78,88,60,94,82,73,97,86,91,93]})

print(df.head())
print(df.describe())

df["Experience_Group"]=pd.cut(df["Experience"],[0,3,7,20],
                              labels=["Low","Medium","High"])

fig,ax=plt.subplots(2,3,figsize=(14,8))
df["Department"].value_counts().plot.bar(ax=ax[0,0],title="Employees by Department")
df.groupby("Department")["Productivity_Score"].mean().plot.bar(ax=ax[0,1],title="Average Productivity by Department")
ax[0,2].scatter(df["Working_Hours"],df["Productivity_Score"])
ax[0,2].set(title="Working Hours vs Productivity",xlabel="Working Hours",ylabel="Productivity")
ax[1,0].scatter(df["Training_Hours"],df["Productivity_Score"])
ax[1,0].set(title="Training Hours vs Productivity",xlabel="Training Hours",ylabel="Productivity")
ax[1,1].hist(df["Productivity_Score"],bins=5,edgecolor="black")
ax[1,1].set(title="Productivity Distribution",xlabel="Score",ylabel="Employees")
df.groupby("Experience_Group",observed=True)["Productivity_Score"].mean().plot.bar(ax=ax[1,2],title="Productivity by Experience")
for a in ax.flat: a.grid(alpha=.25)
plt.tight_layout()
plt.savefig("employee_productivity_dashboard.png",dpi=160)
plt.show()

print("\nAverage productivity by experience:")
print(df.groupby("Experience_Group",observed=True)["Productivity_Score"].mean())
```

Output

Google Colab Output — Employee Productivity Dashboard



Higher experience and training correspond to higher productivity. Employees with more working hours generally show higher scores. The high-experience group has the highest average productivity.

4. Employee Productivity Pattern Analysis

A weather department records Date, Temperature, Humidity, Rainfall and Wind_Speed. Develop a Weather Analysis Dashboard using Python, Pandas and Matplotlib that imports and inspects the dataset, converts Date, shows temperature and rainfall trends, plots temperature versus humidity, displays temperature distribution and rainfall frequency, arranges at least four visualizations in one dashboard, formats the plots, saves the dashboard, and identifies at least three weather patterns.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Date":pd.date_range("2026-06-01",periods=14),
    "Temperature":[32,34,31,29,30,35,36,33,28,30,31,34,35,32],
    "Humidity":[65,60,72,80,78,55,52,62,85,76,73,61,57,66],
    "Rainfall":[2,0,8,15,12,0,0,3,20,10,6,0,0,4],
    "Wind_Speed":[12,15,10,8,9,18,20,14,7,9,11,16,17,13]}

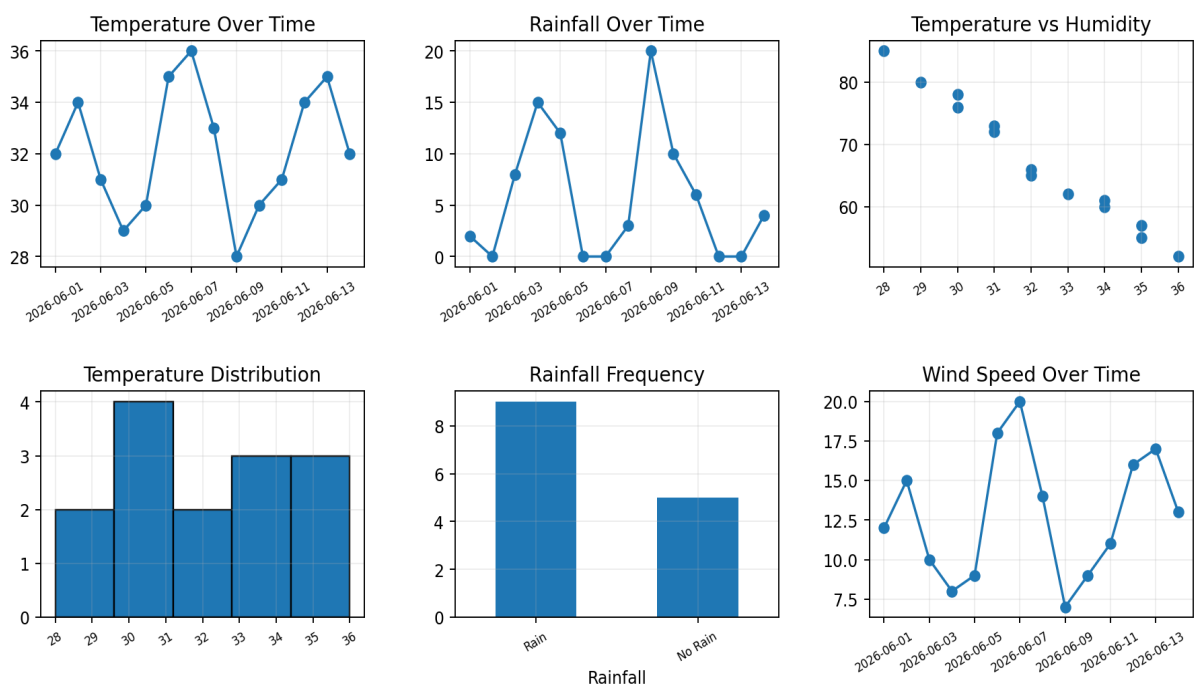
df["Date"]=pd.to_datetime(df["Date"])
print(df.head())
print(df.describe())

fig,ax=plt.subplots(2,3,figsize=(14,8))
ax[0,0].plot(df["Date"],df["Temperature"],marker="o",label="Temperature")
ax[0,0].set(title="Temperature Over Time",ylabel="°C")
ax[0,1].plot(df["Date"],df["Rainfall"],marker="o",label="Rainfall")
ax[0,1].set(title="Rainfall Over Time",ylabel="Rainfall")
ax[0,2].scatter(df["Temperature"],df["Humidity"])
ax[0,2].set(title="Temperature vs Humidity",xlabel="Temperature",ylabel="Humidity")
ax[1,0].hist(df["Temperature"],bins=5,edgecolor="black")
ax[1,0].set(title="Temperature Distribution",xlabel="Temperature",ylabel="Days")
rain_counts=(df["Rainfall"]>0).value_counts()
rain_counts.index=["No Rain","Rain"]
rain_counts.plot.bar(ax=ax[1,1],title="Rainfall Frequency")
ax[1,2].plot(df["Date"],df["Wind_Speed"],marker="o")
ax[1,2].set(title="Wind Speed Over Time",ylabel="Wind Speed")
for a in ax.flat:
    a.grid(alpha=.25)
    a.tick_params(axis="x",rotation=30)
plt.tight_layout()
plt.savefig("weather_dashboard.png",dpi=160)
plt.show()

print("\nPatterns:")
print("1. Higher temperatures generally occur with lower humidity.")
print("2. Rainy days have lower temperatures in this sample.")
print("3. Wind speed is higher on hot, dry days.")
```

Output

Google Colab Output — Weather Monitoring Dashboard



Hotter days generally have lower humidity.
Rainy days are cooler in this sample.
Wind speed tends to be higher on hot, dry days.

5. E-Commerce Dataset: Discovering Hidden Purchase Patterns

A retail store maintains Date, Customer_Age, Gender, Product_Category, Quantity, Sales and Payment_Mode. Develop a Retail Sales and Customer Analysis Dashboard that imports and explores the data, shows sales by category, transaction frequency by payment mode, customer-age distribution, sales over time, quantity versus sales, sales by age group, arranges the visualizations using subplots, applies suitable formatting, saves the dashboard using plt.savefig(), and summarizes the major customer and sales patterns.

Code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Date":pd.date_range("2026-01-01",periods=12,freq="MS"),
    "Customer_Age":[21,35,29,45,38,24,31,52,42,27,34,48],
    "Gender":["M","F","M","F","M","F","M","F","M","F","M","F"],
    "Product_Category":["Electronics","Clothing","Electronics","Furniture","Clothing","Clothing","Electronics","Furniture","Clothing","Electronics","Furniture","Clothing"],
    "Quantity":[2,4,1,2,5,3,2,1,4,3,2,5],
    "Sales":[2500,3200,1800,7000,4500,2100,3000,5200,4100,3600,6800,5000],
    "Payment_Mode":["UPI","Card","UPI","Cash","Card","UPI","Card","Cash","UPI","Card","UPI","Cash"]})

print(df.head())
print(df.describe())

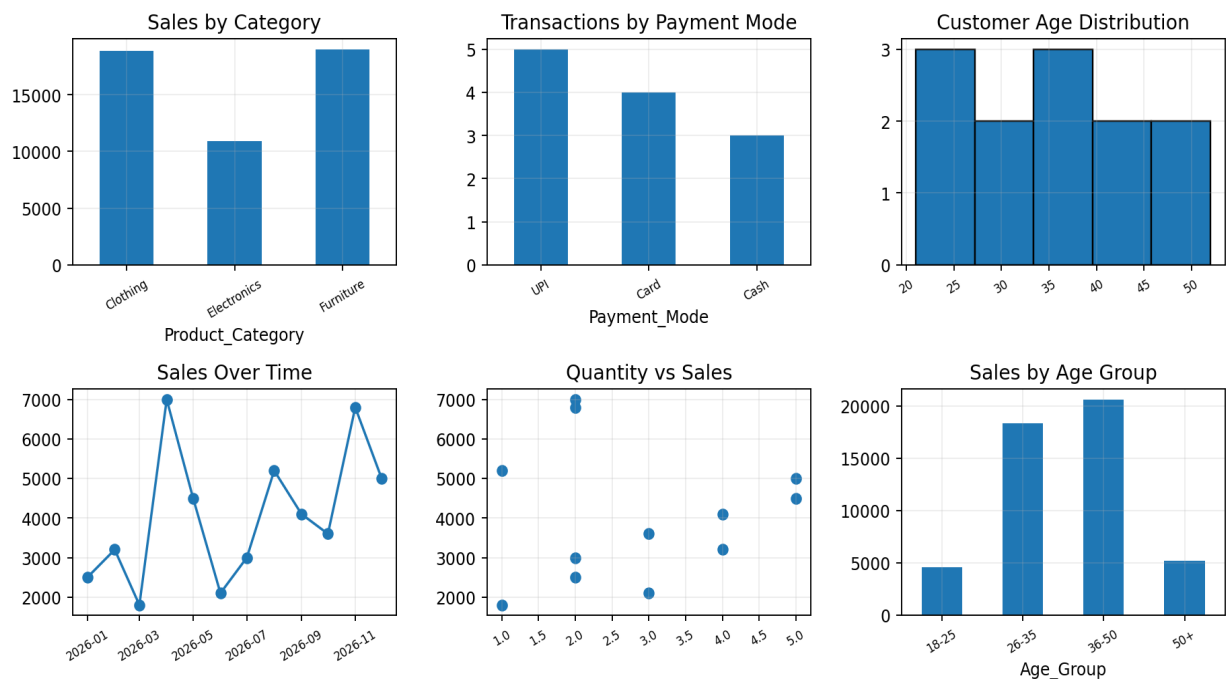
df["Age_Group"]=pd.cut(df["Customer_Age"],[0,25,35,50,100],
    labels=["18-25","26-35","36-50","50+"])

fig,ax=plt.subplots(2,3,figsize=(14,8))
df.groupby("Product_Category")["Sales"].sum().plot.bar(ax=ax[0,0],title="Sales by Product Category")
df["Payment_Mode"].value_counts().plot.bar(ax=ax[0,1],title="Transactions by Payment Mode")
ax[0,2].hist(df["Customer_Age"],bins=5,edgecolor="black")
ax[0,2].set(title="Customer Age Distribution",xlabel="Age",ylabel="Customers")
ax[1,0].plot(df["Date"],df["Sales"],marker="o")
ax[1,0].set(title="Sales Over Time",ylabel="Sales")
ax[1,1].scatter(df["Quantity"],df["Sales"])
ax[1,1].set(title="Quantity vs Sales",xlabel="Quantity",ylabel="Sales")
df.groupby("Age_Group",observed=True)["Sales"].sum().plot.bar(ax=ax[1,2],title="Sales by Age Group")
for a in ax.flat:
    a.grid(alpha=.25)
    a.tick_params(axis="x",rotation=30)
plt.tight_layout()
plt.savefig("retail_dashboard.png",dpi=160)
plt.show()

print("\nPatterns:")
print("1. Furniture has the highest sales in this sample.")
print("2. UPI and Card are the most frequent payment modes.")
print("3. Higher quantities generally correspond to higher sales.")
print("4. The 36-50 age group contributes strongly to sales.")
```

Output

Google Colab Output — Retail Sales & Customer Dashboard



Furniture has the highest sales in this sample.
UPI and Card are the most frequent payment modes.
Higher quantities generally correspond to higher sales.
The 36-50 age group contributes strongly to sales.